# One-Variable Statistics, Distributions & Margin of Error

ClipSAT Printable Practice Suite (clipsat.org) • Topic 9.01 • 20 SPRs + 25 MCQs • Difficulty Progression • Official Digital SAT Curriculum

**ANSWER KEY — COMPLETE WORKED SOLUTIONS**

## Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

**1** [Foundational] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

A data set has values 4, 7, 9, 11, 14. What is the median?

**Answer:** 9. Ordered: 4, 7, 9, 11, 14. Middle value = 9.

**2** [Foundational] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the range of the data set: $84, 37, 93, 17, 57, 98$?

**Answer:** 81. Applying One-Variable Statistics, Distributions & Margin of Error principles, Range $= \text{max} - \text{min} = 98 - 17 = 81$ Therefore, the verified correct option is 81.

**3** [Foundational] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the range of the data set: $50, 23, 12, 96, 77, 47$?

**Answer:** 84. Applying One-Variable Statistics, Distributions & Margin of Error principles, Range $= \text{max} - \text{min} = 96 - 12 = 84$ Therefore, the verified correct option is 84.

**4** [Foundational] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the mean of the data set: $17, 46, 37, 12, 20, 48$?

**Answer:** 30. Mean $= \frac{17 + 46 + 37 + 12 + 20 + 48}{6} = \frac{180}{6} = 30$

**5** [Foundational] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the mean of the data set: $30, 35, 35, 16, 49$?

**Answer:** 33. Mean $= \frac{30 + 35 + 35 + 16 + 49}{5} = \frac{165}{5} = 33$

**6** [Foundational] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the range of the data set: $31, 65, 66, 28, 29$?

**Answer:** 38. Applying One-Variable Statistics, Distributions & Margin of Error principles, Range $= \text{max} - \text{min} = 66 - 28 = 38$ Therefore, the verified correct option is 38.

**7** [Foundational] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the range of the data set: $62, 67, 97, 6, 1$?

**Answer:** 96. Applying One-Variable Statistics, Distributions & Margin of Error principles, Range $= \text{max} - \text{min} = 97 - 1 = 96$ Therefore, the verified correct option is 96.

© ClipSAT — clipsat.org | All Rights Reserved

## Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

**8** [Intermediate] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The table shows the number of pets owned by each student in a class. What is the mean number of pets per student?

**Answer:** 2.6. Total pets: $0(1) + 1(1) + 2(1) + 3(5) + 4(2) = 26$; students: 10. Mean $= \frac{26}{10} = 2.6$.

**9** [Intermediate] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a survey, 40% of 200 students prefer math. How many students prefer math?

**Answer:** 80. Multiply the percentage by the total number of students: $0.40 \times 200 = 80$ students.

**10** [Intermediate] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the mean of the data set: $20, 7, 15, 14, 14$?

**Answer:** 14. Mean $= \frac{20+7+15+14+14}{5} = \frac{70}{5} = 14$

**11** [Intermediate] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the mode of the data set: $44, 16, 10, 44, 45, 9, 12$?

**Answer:** 44. The value 44 appears twice, more often than any other value, so it is the mode.

**12** [Intermediate] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the mode of the data set: $33, 34, 27, 3, 4, 34$?

**Answer:** 34. The value 34 appears twice, more often than any other value, so it is the mode.

**13** [Intermediate] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The table shows test scores for two classes. What is the positive difference between the medians?

**Answer:** 1. Median A = 81; Median B = 82. Difference = 1.

**14** [Intermediate] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The mean of 4 numbers is 36. When one more number, 31, is added to the set, what is the new mean of all 5 numbers?

**Answer:** 35. Original sum $= 36 \times 4 = 144$. New sum $= 144 + 31 = 175$. New mean $= \frac{175}{5} = 35$

## Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

© ClipSAT — clipsat.org | All Rights Reserved

**15** [Challenging] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The mean of 5 numbers is 12. Four of them are 8, 10, 14, 16. What is the fifth?

**Answer:** 12. Applying One-Variable Statistics, Distributions & Margin of Error principles, Sum $= 60$. Fifth $= 60 - 48 = 12$ Therefore, the verified correct option is 12.

**16** [Challenging] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The range of the data set {3, 7, 2, 9, 5, 8} is

**Answer:** 7. The range is the maximum value minus the minimum value: $\text{Maximum} = 9$, $\text{Minimum} = 2$, so $\text{Range} = 9 - 2 = 7$.

**17** [Challenging] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the mean of the data set: $1, 28, 23, 49, 45, 52$?

**Answer:** 33. Mean $= \frac{1+28+23+49+45+52}{6} = \frac{198}{6} = 33$

**18** [Challenging] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the median of the data set: $4, 17, 53, 5, 64$?

**Answer:** 17. Ordered: $4, 5, 17, 53, 64$. The middle value is 17.

**19** [Challenging] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the mean of the data set: $7, 40, 31, 18$?

**Answer:** 24. Mean $= \frac{7+40+31+18}{4} = \frac{96}{4} = 24$

**20** [Challenging] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the median of the data set: $45, 41, 59, 13, 65, 54, 68$?

**Answer:** 54. Ordered: $13, 41, 45, 54, 59, 65, 68$. The middle value is 54.

## Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

**21** [Foundational] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the mode of the data set: $20, 19, 45, 43, 20, 12$?

**a.** 20 ✓

**b.** 27

**c.** 28

**d.** 11

The value 20 appears twice, more often than any other value, so it is the mode.

© ClipSAT — clipsat.org | All Rights Reserved

**22** [Foundational] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the median of the following list of numbers: $3, 7, 12, 15, 19, 24, 30$?

**a.** 12

**b.** 15 ✓

**c.** 17

**d.** 19

The ordered list has 7 values. The middle (4th) value is 15.

**23** [Foundational] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The mean of four numbers is 18. Three of the numbers are 12, 16, and 22. What is the fourth number?

**a.** 20

**b.** 22 ✓

**c.** 24

**d.** 26

Total sum = $4 \times 18 = 72$. Sum of three = $12 + 16 + 22 = 50$. Fourth number = $72 - 50 = 22$.

**24** [Foundational] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The heights of 5 basketball players in inches are 74, 76, 78, 80, and 82. What is the mean height?

**a.** 76 in

**b.** **78 in** ✓

**c.** 80 in

**d.** 82 in

Sum = $390$. Mean = $390/5 = 78$ inches.

**25** [Foundational] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

A scientist measured the weekly rainfall in millimeters over seven weeks: 2, 5, 8, 11, 14, 17, and 20. What was the median weekly rainfall?

**a.** 8

**b.** 10

**c.** 11 ✓

**d.** 14

The middle value in this ordered list of 7 numbers is 11.

© ClipSAT — clipsat.org | All Rights Reserved

**26** [Foundational] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the range of the set: $5, 5, 5, 5, 5$?

**a.** $0$ ✓ **b.** $1$

**c.** $5$ **d.** $25$

Because all elements in the set are identical, the maximum is 5 and the minimum is 5. Thus the range is $5 - 5 = 0$.

**27** [Foundational] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the mean of the numbers: $10, 20, 30, 40, 50, 60$?

**a.** $30$ **b.** $35$ ✓

**c.** $40$ **d.** $45$

Sum = 210. Count = 6. Mean = $210/6 = 35$.

**28** [Foundational] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

A data set has minimum value 24 and range 36. What is the maximum value?

**a.** $50$ **b.** $56$

**c.** $60$ ✓ **d.** $64$

Maximum = Minimum + Range = $24 + 36 = 60$.

## Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

**29** [Intermediate] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The table shows the number of pets owned by each student in a class. What is the mean number of pets per student?

**a.** $8.4$ **b.** $1.5$ ✓

**c.** $5.6$ **d.** $2$

Total pets: $0(9) + 1(8) + 2(4) + 3(2) + 4(5) = 42$; students: 28. Mean $= \frac{42}{28} = 1.5$.

© ClipSAT — clipsat.org | All Rights Reserved

**30** [Intermediate] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The table shows the number of pets owned by each student in a class. What is the mean number of pets per student?

**a.** 5.2

**b.** 10.4

**c.** 4

**d.** 2 ✓

Total pets: $0(6) + 1(6) + 2(4) + 3(2) + 4(8) = 52$; students: 26. Mean $= \frac{52}{26} = 2$.

**31** [Intermediate] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

A biologist recorded the weights of six laboratory mice in grams: 14, 28, 19, 45, 12, and 33. What is the range of these weights?

**a.** 27

**b.** 31

**c.** 33 ✓

**d.** 45

Range = Maximum - Minimum = $45 - 12 = 33$.

**32** [Intermediate] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the median of the following even-sized data set: $4, 9, 11, 15, 18, 24$?

**a.** 11

**b.** 12

**c.** 13 ✓

**d.** 15

The middle two numbers are 11 and 15. The median is their average: $(11 + 15)/2 = 13$.

**33** [Intermediate] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the range of the scores: $85, 92, 78, 95, 88$?

**a.** 14

**b.** 15

**c.** 17 ✓

**d.** 18

Applying One-Variable Statistics, Distributions & Margin of Error principles, Range = $95 - 78 = 17$. Therefore, the verified correct option is 17.

© ClipSAT — clipsat.org | All Rights Reserved

**34** [Intermediate] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

A frequency table shows the number of siblings reported by 20 students. The results are shown in the table. What is the mode of the number of siblings?

**a.** 0 **b.** 1 ✓

**c.** 2 **d.** 8

The sibling count with the highest frequency (8 students) is 1.

**35** [Intermediate] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a class of 25 students, the median test score is 84. If 12 students scored above 84, how many students scored at or below 84?

**a.** 12 **b.** 13 ✓

**c.** 14 **d.** 15

Total students = 25. If 12 scored above 84, then $25 - 12 = 13$ scored at or below 84.

**36** [Intermediate] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The temperature in degrees Celsius over 5 days was: $-2, 0, 4, 7, 11$. What was the median temperature?

**a.** 0°C **b.** 2°C

**c.** **4°C** ✓ **d.** 7°C

Ordered list has middle value 4°C.

**37** [Intermediate] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

If the mean of three numbers is 10, what is the sum of the three numbers?

**a.** 10 **b.** 20

**c.** 30 ✓ **d.** 40

Applying One-Variable Statistics, Distributions & Margin of Error principles, Sum = $3 \times 10 = 30$. Therefore, the verified correct option is 30.

## Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

© ClipSAT — clipsat.org | All Rights Reserved

**38** [Challenging] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The numbers of books read by five students last month are: 6, 10, 14, 18, and 22. What is the mean number of books read?

**a.** 12

**b.** 14 ✓

**c.** 16

**d.** 18

Sum = $6 + 10 + 14 + 18 + 22 = 70$. Count = 5. Mean = $70/5 = 14$.

**39** [Challenging] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

A teacher recorded the number of questions answered correctly by eight students: 4, 8, 8, 12, 15, 15, 15, and 20. What is the mode of these scores?

**a.** 8

**b.** 12

**c.** 15 ✓

**d.** 20

The value 15 appears most frequently (3 times).

**40** [Challenging] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

A survey asked 150 students how many hours per week they read. The results are shown in the table. How many students read at least 6 hours per week?

**a.** 40

**b.** 50

**c.** 55 ✓

**d.** 60

At least 6 hours includes '6–8' (40) and '9+' (15): $40 + 15 = 55$.

**41** [Challenging] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

If five quiz scores have a sum of 420, what is the mean score?

**a.** 80

**b.** 82

**c.** 84 ✓

**d.** 86

Applying One-Variable Statistics, Distributions & Margin of Error principles, Mean = $420/5 = 84$. Therefore, the verified correct option is 84.

© ClipSAT — clipsat.org | All Rights Reserved

**42** [Challenging] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The mean of five test scores is 82. If four of the scores are 80, 85, 75, and 90, what is the fifth score?

**a.** 78

**b.** 80 ✓

**c.** 82

**d.** 84

Total needed = $5 \times 82 = 410$. Current sum = $80 + 85 + 75 + 90 = 330$. Fifth score = $410 - 330 = 80$.

**43** [Challenging] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

Find the median of the data set: $18, 22, 14, 30, 26$.

**a.** 18

**b.** 20

**c.** 22 ✓

**d.** 26

Order the data: 14, 18, 22, 26, 30. The middle value is 22.

**44** [Challenging] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the mode of the values: $12, 15, 18, 15, 21, 15, 18$?

**a.** 12

**b.** 15 ✓

**c.** 18

**d.** 21

15 appears 3 times, which is more than any other value.

**45** [Challenging] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the median of: $50, 60, 70, 80$?

**a.** 60

**b.** 65 ✓

**c.** 70

**d.** 75

Applying One-Variable Statistics, Distributions & Margin of Error principles, Average of 60 and 70 is 65.

Therefore, the verified correct option is 65.

© ClipSAT — clipsat.org | All Rights Reserved